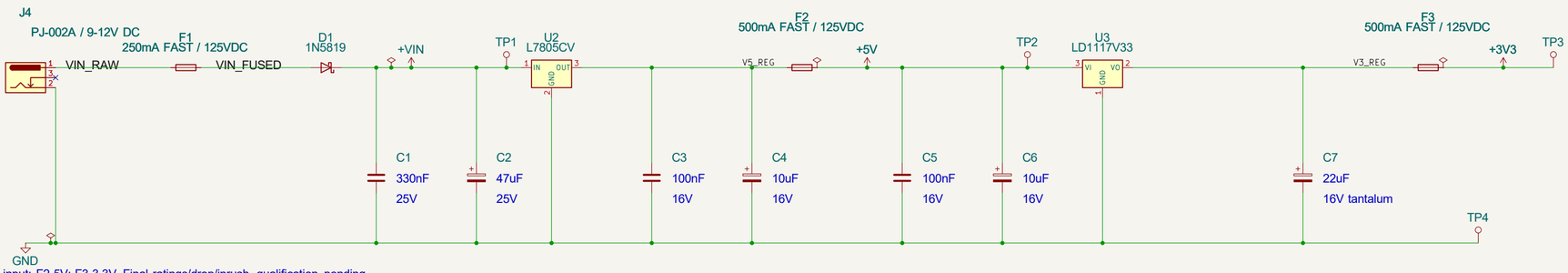
WEATHER GADGET — LCD DISPLAY SUBSYSTEM

Zander Boward | Team 202 | Revision H | 30 October 2026 | CN0293 LCD1602

ONE A2 SHEET | 5 V PIC / LCD logic | 3.3 V LCD backlight | Three UART links

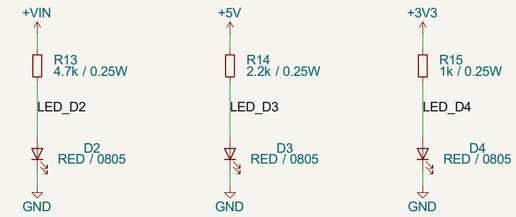
9–12 V INPUT / LINEAR REGULATORS



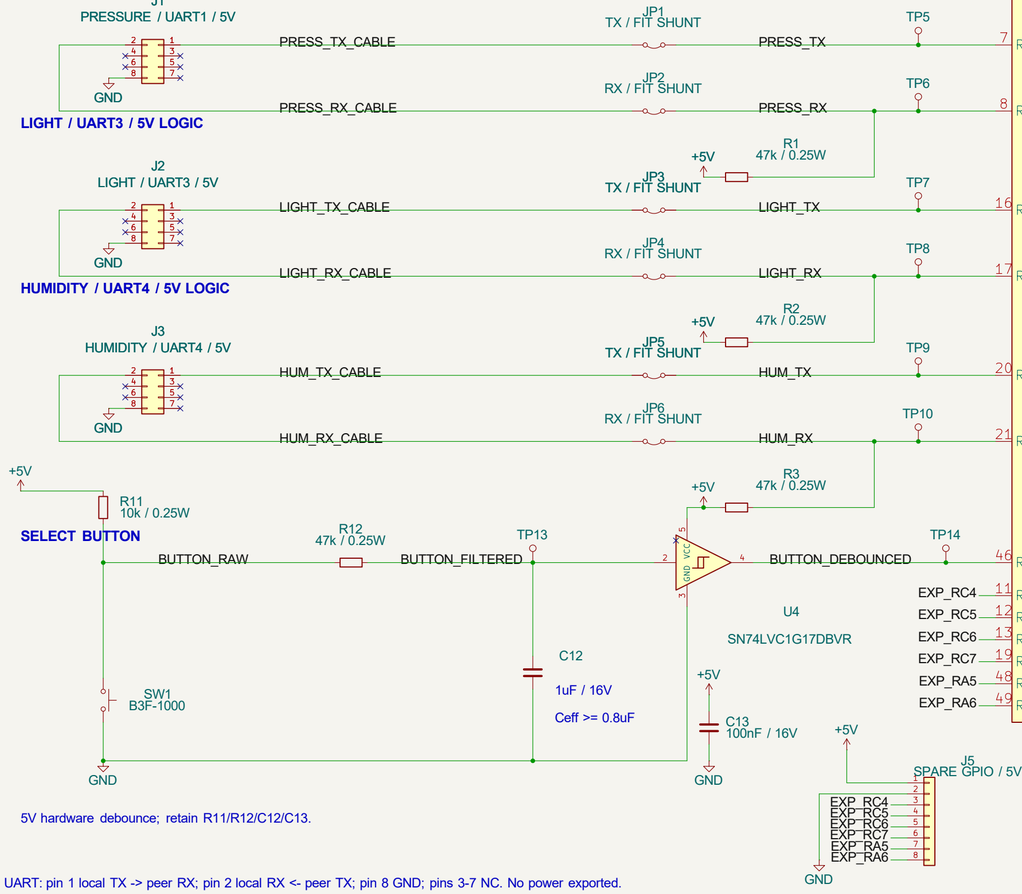
F1 input; F2 5V; F3 3.3V. Final ratings/drop/inrush qualification pending.

U2 tab = GND; U3 tab = OUT. Do not join tabs. C7 stability requires a bench test.

POWER INDICATORS



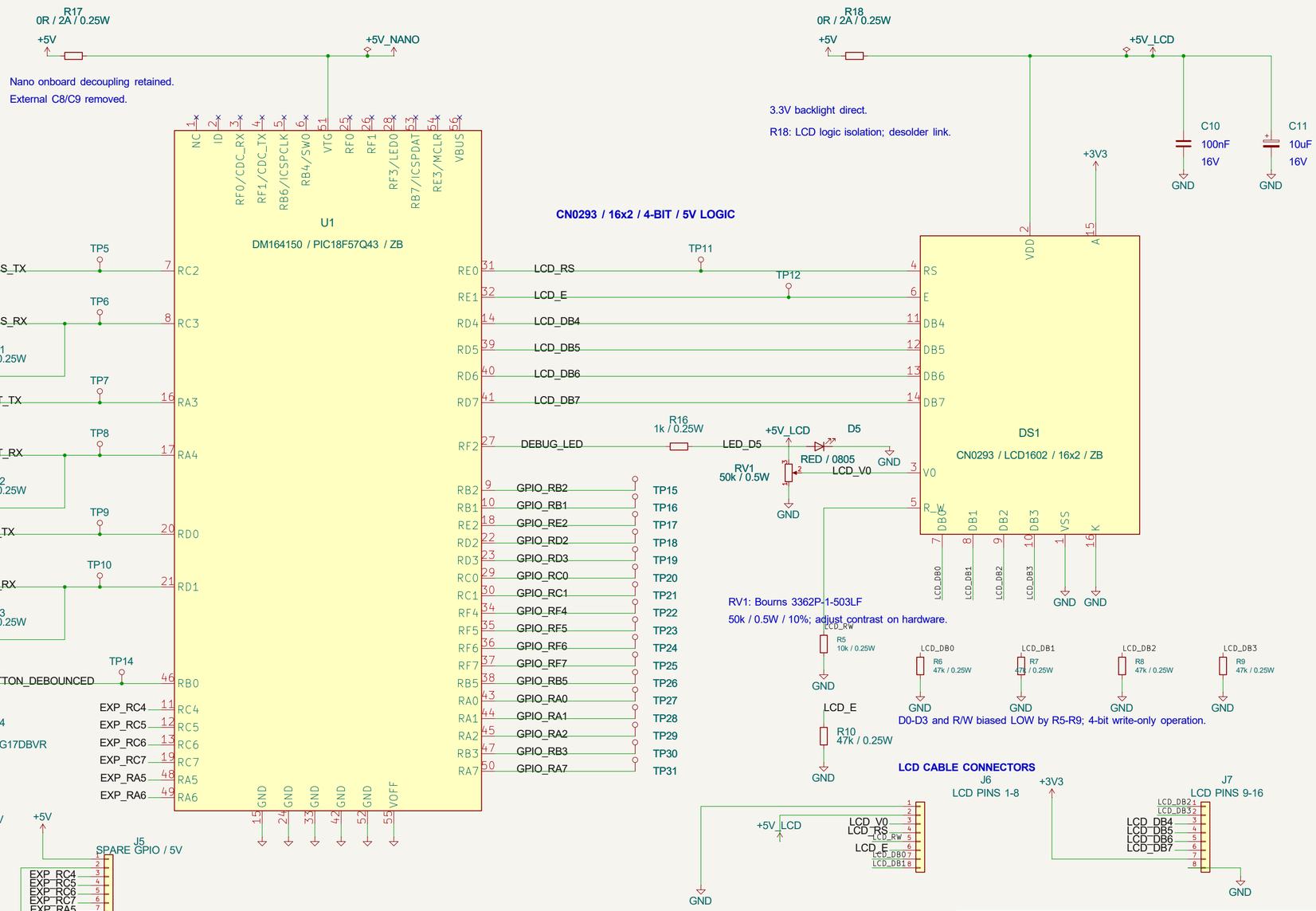
PRESSURE / UART1 / 5V LOGIC



U1 / DM164150 / PIC18F57Q43 / ZB

CN0293 / 16x2 / 4-BIT / 5V LOGIC

+5V_NANO / +5V_LCD



LCD CABLE CONNECTORS

SPARE GPIO / 5V

UART: pin 1 local TX -> peer RX; pin 2 local RX <- peer TX; pin 8 GND; pins 3-7 NC. No power exported.

Verify peer 5 V compatibility and cable continuity. VTG 51 = 5 V; VOFF 55 = GND; VBUS 56 NC.

Fit shunts. Power OFF before moving them; signal pins can backfeed unpowered boards. Hardware qualification remains pending.

Nano current allocation provisional; verify rail loads, F1 inrush, regulator temperature, C7 stability and LCD current.

Original hardware and team interface qualification pending
CN0293; 5V PIC; one input fuse; 50k contrast trimmer
Team 202 | Zander Boward

Sheet: /
File: LCD-Board-Connected.kicad_sch

Title: Weather Gadget — LCD Display Subsystem

Size: A2	Date: 2026-10-30	**Rev: H**
KiCad E.D.A. 10.0.7		Id: 1/1